

CHAPTER 2 CRYPTOGRAPHIC TOOLS

- 2.1** Define the notion of collision resistance for hash functions.
- 2.2** Consider an application that requires an encryption and MAC algorithm be implemented on a processor with a small amount of non-volatile memory. The only cryptographic algorithm that the processor can compute is Triple-DES. But you have space for one 168-bit key. Describe how it is possible to both encrypt and MAC using only a single key. Justify your answer and state any assumptions you use.
- 2.3** We consider a banking application, where messages m of the form `fromAccount, toAccount, amount` are sent within the bank network, with the meaning that `amount` dollars should be transferred from `fromAccount` to `toAccount`. Each message consists of three blocks, with each block holding one of the three parameters. Messages are encrypted using AES in Counter mode, i.e.

$$K_j = E(K, T_j)$$
$$C_j = M_j \oplus K_j$$

Each of the three parts of a message is sixteen characters, i.e. one block, so messages consist of three blocks.

- a.** The adversary has an account in the bank and can intercept and change messages. Imagine now that he knows the `toAccount` for a particular message $m = C_1C_2C_3$. Explain how he can modify the message so that the amount is transferred to his own account
- b.** Explain how the use of a MAC would prevent this attack.
- c.** Above, $E(K, M)$ denotes using block cipher E with key K on message M . It is possible to define a cipher using similar ideas, but using a hash function instead. Describe how to do it, including how to decrypt.